Lecture 17: A Person in the World of People: Self and Other, Part II; Some Mysteries: Sleep, Dreams, and Laughter

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-17 (CC BY-NC-SA 3.0).

Before you start

The first half finishes social psychology: how we form impressions of others and how we think about groups. The second half looks at three "mysteries" that psychology still cannot fully explain: sleep, dreams and laughter.

Learning goals

After this lecture you should be able to:

- Explain why first impressions matter and what "thin slices" research shows.
- Describe the good and bad sides of stereotypes, including stereotype threat.
- Explain how implicit attitudes are tested and why they matter.
- Compare slow-wave and REM sleep, and explain one theory of dreams and one of laughter.

Key ideas

1. Introduction

Professor Bloom reviews the last lecture: biases about the self, the fundamental attribution error, and why we like people (proximity, similarity, attractiveness). The Matthew effect means that good things tend to add up.

2. First and Fast: How We Form Impressions of Others

- **First impressions** matter because of the confirmation bias: we interpret later information to fit what we first believed. In Kelley's study, students read a bio describing a guest speaker as very warm or as rather cold, and this biased their impressions of him.
- **Thin slices:** we form impressions very fast. Rosenthal and Ambady found that after five-second clips of professors, people were pretty good at predicting their teaching evaluations.
- **Pygmalion effect:** if I believe you have a trait, this may cause you to behave as if you have it. Rosenthal and Jacobson told teachers that some children, really chosen at random, would show a jump in IQ. They did. "This isn't magic": teachers give such children more attention.

3. Positive Uses and Negative Effects of Stereotypes

- Psychologists use **stereotype** in a broad sense: information about categories and what is typical of them. This is essential: we expect an apple to be edible. Stereotypes about people can be positive,

and they are often accurate.

- But they are not always accurate. Because of the confirmation bias, we give extra weight to cases that fit. Media data may not be representative. In classical music, blind auditions led to far more women being hired.
- There are moral problems too. Some people hold that even a correct stereotype is immoral to apply to people ("profiling"), yet we accept higher car insurance for young drivers.
- **Stereotype threat** (Claude Steele): if your group has a negative stereotype in an area, a reminder of your group before a test lowers your performance, for example black people or women on a math test.
- There are three levels: public, private and unconscious. The share of people who say they would vote for an African American president rose from about half to almost one hundred percent. But when black faces were flashed too fast to see, people more often completed "hos-" as "hostile"; after white faces, as "hospital".

4. Implicit Attitudes

- Mahzarin Banaji helped develop the **Implicit Association Test** (the lecture calls it the "implicit attitudes test"), which looks for unconscious associations. You sort faces and good or bad words quickly. People are slower when white faces go with bad words and African American faces with good words than the other way round.
- This half-second difference can matter in split-second choices (studies of police officers deciding whom to shoot) and in judgment calls.
- Jack Dovidio: with strong recommendations, African American and white candidates were hired equally. With so-so recommendations, the white candidate was more likely to be hired, in 1989, 1999 and recently.
- Trish Devine's **automaticity theory:** everyone holds stereotypes, which are activated automatically. We can consciously override them, but it takes work.

5. Question and Answer on Stereotypes

Group-level fixes include blind reviewing in journals and special efforts to get names from disadvantaged groups in job searches. African Americans show much the same effect on the test as white Americans, so it is not only favoritism for one's own group.

6. The Minor Mystery of Sleep

- Sleep is a motivation, like food. An **EEG** (electrodes on the scalp) shows brain waves: beta waves when focused, alpha waves when awake but not attentive, delta waves in deep sleep.
- We go from stage I down to stage IV, back up, and then into **REM sleep** (rapid eye movement). A good night has four to five cycles.
- In **slow-wave ("quiet") sleep**, you are hard to wake up. In REM sleep, the brain looks awake, the body is paralyzed except for the eyes, and dreams occur.
- Why sleep? Probably to restore the body, like stopping a car's engine before you repair it. Another view: to save energy and keep you out of trouble at night.

7. The Greater Mystery of Dreams

- **Real dreams** (strange stories, in REM sleep) differ from **sleep thought** (repeated thoughts like "did I take the garbage out?", in slow-wave sleep).
- Everybody dreams three to four times a night, but dream memories fade quickly.
- From 50,000 dream reports collected by Hill: most dreams are on the negative side, and men have more aggressive dreams than women. The most common dream is being chased.
- Freudian theories have little support. A popular theory: dreams are a side effect of **memory consolidation**, as the brain "backs itself up". From this view, dreams are **epiphenomena**: byproducts with no adaptive function.

8. The True Mystery of Laughter

- A textbook says we laugh at incongruity (a mismatch between what we expect and what happens) unless it is frightening. Bloom says this is "about as wrong as anything can be": it does not explain the loud noise, and much laughter has no incongruity.
- Robert Provine observed when people laughed, for example in shopping malls. Only ten percent of the comments before laughter were humorous ("I'll see you guys later" caused laughter). So laughter is separate from jokes.
- Laughter is universal, appears early, is loud (so it is social and communicative) and is contagious.
- One theory: laughter signals **mock aggression** (tickling is a mock attack) and collective (group) aggression, and it shows solidarity.
- Ingredients of humor: a target who suffers some harm; harm not so serious that it causes fear, grief or pity (a banana peel, not a split head); and surprise. We cannot usually tickle ourselves because there is no surprise and no mock aggression.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
confirmation bias	interpreting new information so it supports what we already believe	the "warm" or "cold" guest speaker
thin slices	very short views of a person that still give an impression	five-second clips of professors
Pygmalion effect	others' expectations make a person act in the expected way	Rosenthal and Jacobson's IQ study
stereotype	information about what is typical of a category	"this dog probably barks"
stereotype threat	a reminder of a negative group stereotype lowers performance	Claude Steele's math tests
implicit attitudes	unconscious associations we do not know we have	the Implicit Association Test
automaticity theory	stereotypes come to mind automatically and must be consciously overridden	Trish Devine's theory

Term	Meaning in simple English	Example from the lecture
EEG	a recording of electrical activity in the brain	beta, alpha and delta waves
REM sleep	sleep with rapid eye movements, an active brain and dreams	"a monkey was eating my grandmother"
slow-wave sleep	deep, "quiet" sleep, when you are hard to wake up	"did I take the garbage out?"
epiphenomena	byproducts of another system, with no function of their own	dreams as side effects of memory consolidation
mock aggression	a pretend attack that is not real	tickling

Check your understanding

1. Why do first impressions matter so much? Give an example.
2. What are "thin slices"? Describe the study on professors.
3. Describe Rosenthal and Jacobson's study. What does it show?
4. Give one reason why stereotypes can be useful and one problem with them.
5. What is stereotype threat?
6. What do Jack Dovidio's hiring studies show?
7. Compare slow-wave sleep and REM sleep.
8. A friend says, "People laugh because something is funny." Use Provine's research to reply.

Answer guide

1. Because of the confirmation bias, we interpret later information to fit it, as with the "warm" or "cold" speaker.
2. Very brief views of a person. After five-second clips, people predicted professors' teaching evaluations pretty well.
3. Randomly chosen children whom teachers expected to improve did show an IQ jump. Expectations matter.
4. Useful: they let us make generalizations that are often correct. Problem: they can be wrong, as blind auditions showed.
5. A reminder of a negative stereotype about your group before a task lowers your performance.
6. With so-so recommendations, white candidates were more likely to be hired, so implicit stereotypes affect judgment calls.
7. In slow-wave sleep you are deeply asleep and hard to wake. In REM sleep the brain looks awake, the body is paralyzed and dreams occur.
8. Provine found that only ten percent of the comments before laughter were humorous.

English practice

Useful phrases from the lecture

- "It turns out that ..." - introduces a surprising finding.
- "The idea is ..." - introduces the main point of a theory.
- "This isn't magic." - says a surprising result has a simple explanation.
- "So, the general takeaway message here is that ..." - gives the main lesson.
- "This is a fair effort." - polite praise before a criticism.

Speaking task (with your AI tutor)

- Explain implicit attitudes in 4-6 sentences. Use "Implicit Association Test", "automatically" and "override".

Writing task

- Write one paragraph (80-120 words): Bloom asked for "an interesting testable idea about either dreams or laughter". Describe one theory from the lecture and suggest a simple study to test it.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 17. Ask me questions about first impressions and stereotypes, one at a time."
- "Quiz me on the key vocabulary of Lecture 17."

- "I will explain the stages and types of sleep in my own words. Check my understanding and my English."
- "Ask me why laughter is a mystery and what the mock aggression theory says."

Readings from the original course

- Gray, Peter. Psychology (5th edition), chapter 16

My notes